

1 • Shared evidence: nursery cases

The plant reproduction strategy studio

Name _____ Date _____

Cases are constructed teaching examples using the stated biological processes.

A • Crossed flowers

Pollen from one pea plant reaches another. A male and female gamete fuse before the seed develops.

C • Strawberry runner

A horizontal stem grows from a parent. It roots at a node and forms a daughter plant without gamete fusion.

E • Nursery brief

A fictional nursery wants to preserve a parent mint plant's inherited features. It can use cuttings; growth conditions still need care.

B • Self-fertile flower

A flower receives its own plant's pollen. Male and female gametes fuse before a seed develops.

D • Mint cutting

A piece of a parent stem roots and grows into a new plant. No gamete fusion occurs in this route.

F • Limits card

Clones usually share inherited information, apart from changes such as mutations. Conditions affect appearance. Some plants can make seeds without fertilisation; identify the stated process.

W1. Sort A-D into sexual/asexual and cite the deciding phrase.

W2. Recommend a route for the mint nursery and one condition to control.

W3. Repair the one-parent and identical-clone rules.

2 • Supported route: find the fusion

Use page 1. Word bank: sexual / asexual / gametes / runner.

S1. In case B, gametes fuse. Is this sexual or asexual?

- A. Sexual
- B. Asexual

S2. A strawberry runner grows a daughter plant without fusion. Is this sexual or asexual?

- A. Sexual
- B. Asexual

S3. Complete: Sexual reproduction involves _____ fusing. A _____ can form a new plant without that fusion.

S4. Which method could retain the mint parent's inherited features: cutting or a new cross? Explain.

S5. Give one reason two clones might look different.

3 • Standard route: nursery decision

Use the process evidence, not a visual parent count.

Case	Route	Evidence phrase
A		
B		
C		
D		

N1. Complete the case table.

N2. Explain why case B is sexual despite one parent plant.

N3. Recommend a method for the mint brief. Give a reason and one growth condition to control.

N4. Repair “clones always look identical”.

4 • Stretch route: test the definitions

Avoid all/never claims unsupported by the stated cases.

T1. Compare sexual and asexual reproduction using fusion, inherited information and one named method.

T2. Explain why “two parents versus one” is an unreliable complete definition for plants.

T3. A plant forms seeds without gamete fusion. Which reproduction category fits the stated evidence? Explain.

T4. Evaluate a nursery promise: “Every cutting will look identical and resist every disease.” Give a better claim and one practical limitation.

5 • Exit and reflection

Short reasons matter more than labels alone.

E1. What event defines sexual reproduction?

E2. Describe one asexual plant method.

E3. Can one self-fertile plant reproduce sexually? Explain.

E4. Why might genetically similar clones look different?

Reflection. Which nursery question needs more evidence?
